

# ERRATA & COMMENTS

## COMBINATORIAL RECIPROCITY THEOREMS

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- p. 6 Proof of Proposition 1.1.3: the inequalities should go the other way.
- p. 16 In the last paragraph (“To make the combinatorial reciprocity...”), the maps should be from  $[2]$  into  $[n]$ .
- p. 25 Exercise 1.15 (b): the word “nowhere-zero” needs to be removed.
- p. 32 Exercise 2.5 asserts that the sum on the right stops at the index  $k = |\Pi|$ , but this is a bit crude: the sum actually stops at the length of the longest chain in  $\Pi$ .
- p. 38 Theorem 2.3.2: The exponent should be the length of  $\Pi$  (instead of  $|\Pi|$ ).
- p. 39 The last binomial coefficient on p. 39 should have a  $-1$  at the end instead of the  $+1$ . The same corrections should be made on top of p. 40.
- p. 42 In the last displayed math line, the exponent should be  $\kappa$ , not  $c$ .
- p. 44 In the proof of Theorem 2.4.5, “if  $I = I_S$ ” should be replaced by “if  $I = J_S$ ” in the definition of  $F_-$ .
- p. 44 Just before Theorem 2.4.6, it should say  $a = a_0 \prec a_1 \prec \dots \prec a_k = b$ .
- p. 49 Exercise 2.16: The  $I$  in the exponent should be a  $J$ .
- p. 82 In the proof of Proposition 3.5.2,  $T_{\mathbf{q}}(\mathbf{Q}) = T_{\mathbf{q}-\mathbf{r}}(\mathbf{Q} - \mathbf{r})$  should be  $T_{\mathbf{q}}(\mathbf{Q}) = T_{\mathbf{q}-\mathbf{r}}(\mathbf{Q} - \mathbf{r}) + \mathbf{r}$ .
- p. 90 In the proof of Theorem 3.6.4,  $r(B)$  and  $b(B)$  should be  $r(\mathcal{H})$  and  $b(\mathcal{H})$ .
- p. 95 Exercise 3.2: the second sentence is incomplete. The correct statement is, an irredundant presentation is a set  $I \subseteq [m]$  such that  $\mathbf{Q} = \bigcap_{i \in I} H_i^{\leq}$  but  $\mathbf{Q} \neq \bigcap_{i \in J} H_i^{\leq}$  for any proper subset  $J \subset I$ .
- p. 96 Exercise 3.4(b) should read  $\mathbf{p} + \mathbb{R}_{\geq 0} \mathbf{u} \subseteq \mathbf{Q}$  for all  $\mathbf{p} \in \mathbf{Q}$  and  $\mathbf{u} \in \text{rec}(\mathbf{Q})$ .
- p. 97 Exercise 3.9: here we want to require the set to be closed, not just convex.
- p. 101 Exercise 3.44: On the first two lines of this exercise,  $L$  needs to be replaced by  $L'$  in two occurrences.
- p. 125 The rational function in the middle of the page (just before “This implies, again with (4.6.4)”) should have  $1 - z_1 z_2 z_3^2$  as its last factor in the denominator.
- p. 141 In addition to the note on Theorem 4.2.2 and Cayley’s work on composition, there is an illustrious connection to Vedic poetry; see “The So-Called Fibonacci Numbers in Ancient and Medieval India” by Parmanand Singh, *Historia Mathematica* 12 (1985), 229–244.
- p. 149 Exercise 4.37: the first vector in the second line should be  $(1, 1, 1)$ .
- p. 198 Exercise 5.18: the first inequality should read  $a_0 x \geq b_0$ .
- p. 207 In Theorem 6.2.2, the phrase “crosscut in  $\mathcal{N}$ ” needs to be replaced by “collection of elements in  $\mathcal{N}$  such that every minimal element is uniquely covered”.
- p. 226 The literature contains different (and unfortunately conflicting) definitions of the comajor index.
- p. 240 In the string  $0 = x_u = x_{v_0} < \dots < x_{v_k} = x_u$ , the second  $x_u$  should be  $x_v$ . In the following line,  $u$  should be replaced by  $v$ .
- p. 262 In the last line before Proposition 7.5.9, it should say  $1 \leq k < d$ .

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